

		Option A will give the same resistance.
21	B	Consider a power supply of e.m.f. E and internal resistance r connected to a load of resistance R. $\text{Power output} = P = \frac{V_T^2}{R} = \frac{\left(\frac{R}{R+r}E\right)^2}{R} = \frac{RE^2}{(R+r)^2}$ For P, $24.0 = \frac{R(12)^2}{(R+0)^2}$ $R = 6.0 \, \Omega$ For Q, $13.5 = \frac{(6.0)(12)^2}{(6.0+r)^2}$ $r = 2.0 \, \Omega$

No.	Answer	Solution
22	A	P is a diode which will allow a current to pass through in one direction